

1. If the signal $x[n] = 8 \cos(\frac{\pi}{6}n + \frac{\pi}{2})$ is obtained through the 9.6KHz sampling, then the true frequency of the original analog signal is 800Hz
2. $x[n]$ is a finite length sequence, $x[n] = \{2, 3, 1, 4, -3, -1, 1, 0, 6\}$, $X[k]$ is the 10 points DFT of this sequence, so $X[0] =$ 14
3. A 9 points average filter is described by $y[n] = \frac{1}{9} \sum_{l=0}^{9-1} x[n-l]$, please find out its group delay $\tau(w) =$ 4
4. If a finite length sequence $\{x[n]\} = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, then $x[\langle 4-n \rangle_9] =$ $\{9, 8, 7, 6, 5, 4, 3, 2, 1\}$
5. $\{X_1[k]\}$ is obtained by the 6-point DFT transformation of the sequence $\{x[n]\} = \{1, 2, 2, 3, 3, 0\}$. Apply the IDFT to $\{X_1[k]\}$, we get $\{y_1(n)\} =$ $\{1, 2, 2, 3, 3, 0\}$. $\{X_2[k]\}$ is obtained by the 3-point DFT of $\{x[n]\}$. Apply the IDFT to $\{X_2[k]\}$, we get $\{y_2(n)\} =$ $\{6, 3, 2\}$.
6. Given a 9-point real sequence with DFT values of $X[0] = 23, X[1] = 2.2426 - j, X[4] = -6.3749 + 4.1212j, X[6] = 6.5 + 2.5891j, X[7] = -4.1527 - 0.2645j$ at five points, determine the DFT values at the remaining 2 points: $X[2] =$ $-4.1527 + 0.2645j$, $X[8] =$ $2.2426 + j$.
7. Let the sequence $x[n]$ of length N be padded with $(M-1)N$ zeros, resulting in an MN -point DFT $Y[k]$. Prove that the N -point DFT $X[k]$ of $x[n]$ can be obtained from $Y[k]$ by the following formula.

$$X[k] = Y[Mk], \quad 0 \leq k \leq N-1$$

证明:

$$Y[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/MN}, k = 0, 1, \dots, MN-1$$

$$Y[Mk] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi Mk n/MN} = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N} = X[k], \quad k = 0, 1, \dots, N-1$$

8. A known sequence

$$x(n) = \{-1-j, 3+j, 2+j, 2-j, -8+j, 4-j\} \quad 0 \leq n \leq 5,$$

its FFT is $X[k]$. Write down its circularly conjugate symmetry $x_{cs}(n)$ and

circularly conjugate antisymmetry $x_{ca}(n)$, and use $X[k]$ to indicate their FFT.

解:

$$x(n) = \{-1-j, 3+j, 2+j, 2-j, -8+j, 4-j\} \quad 0 \leq n \leq 5$$

$$x^*_{<-n>_5} = \{-1+j, 4+j, -8-j, 2-j, 2-j, 3-j\} \quad 0 \leq n \leq 5$$

$$x_{cs}(n) = \frac{x(n) + x^*_{<-n>_5}}{2}$$

$$x_{cs}(n) = \left\{ -1, \frac{7+2j}{2}, \frac{-6+5j}{2}, 2, \frac{-6-5j}{2}, \frac{7-2j}{2} \right\}$$

$$x_{ca}(n) = \frac{x(n) - x^*_{<-n>_5}}{2}$$

$$x_{ca}(n) = \frac{x(n) - x^*_{<-n>_5}}{2} = \left\{ -7j, -\frac{1}{2}, \frac{10+9j}{2}, 2j, \frac{-10-9j}{2}, \frac{1}{2} \right\}$$

$$x_{cs}(n) \leftrightarrow X_{re}[k]$$

$$x_{ca}(n) \leftrightarrow jX_{im}[k]$$

9. Let $X[k], 0 \leq k \leq 11$, be a 12-point DFT of a length-12 real sequence $x[n]$ with the first seven samples of $X[k]$ given by

$$X(k) = \{12, -18 - j21, -10 + j4, -6 + j7, 9 + j8, 19 - j16, 39\} \quad 0 \leq k \leq 6$$

- 1) Determine the remaining samples of $X[k]$.
- 2) Evaluate the following functions of $x[n]$ without computing the IDFT of

$$X[k]: \text{(a) } x[6] \quad \text{(b) } \sum_{n=0}^{11} x[n] \quad \text{(c) } \sum_{n=0}^{11} e^{j\pi n/3} x[n]$$

解:

$$X(k) = \{12, -18 - j21, -10 + j4, -6 + j7, 9 + j8, 19 - j16, 39, 19 + j16, 9 - j8, -6 - j7, -10 - j4, -18 + j21\} \quad 0 \leq k \leq 11$$

$$x[6] = \frac{1}{12} \sum_{k=0}^{11} (-1)^k X[k] = 4.9167$$

$$\sum_{n=0}^{11} x[n] = X[0] = 12$$

$$\sum_{n=0}^{11} e^{j\pi n/3} x[n] = \sum_{n=0}^{11} e^{j2\pi 2n/12} x[n] = X[\langle 0-2 \rangle_{12}] = X[10] = -10 - j4$$